

INDUSTRIAL PUMP P-102

Standard Operating Procedure (SOP) & Condition Inspection Manual

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1. Purpose

This procedure provides guidance for safe operation, routine inspection, and maintenance of Industrial Pump P-102. It is intended to support maintenance personnel in identifying visible abnormalities and selecting appropriate inspection actions.

2. Safety

Before intrusive maintenance, isolate the pump from electrical and process energy and follow the site's lockout/tagout procedure. Do not remove guards or open pressurized components while the equipment is operating. AI-assisted observations must be verified by a qualified maintenance professional before shutdown, repair, or restart decisions.

3. Visual Inspection Points

Inspect the following areas during routine condition checks:

- Mechanical seal area: Look for fluid residue, active leakage, staining, or abnormal spray.
- Shaft/coupling area: Check for visible damage, abnormal movement, or misalignment indicators.
- Bearings: Look for overheating indicators, lubricant leakage, unusual noise, or vibration symptoms.
- Pump casing and piping: Check for corrosion, cracks, loose connections, and unusual deposits.
- Base and fasteners: Check for looseness, corrosion, or signs of movement.

4. Leakage Assessment

Minor or intermittent residue around the mechanical seal should trigger closer inspection and documentation. Continuous or significant leakage requires prompt maintenance assessment. If leakage is accompanied by abnormal vibration, temperature, noise, loss of performance, or a safety hazard, stop the equipment according to the approved shutdown procedure and escalate to qualified personnel.

5. Recommended Inspection Sequence

1. Confirm machine identification and operating condition.
2. Review recent maintenance history.
3. Perform external visual inspection.
4. Inspect the mechanical seal and surrounding area if leakage is suspected.
5. Check bearings, coupling, lubrication, piping, and mounting.
6. Compare observations with this SOP and applicable site limits.
7. Document findings and create a maintenance task where required.

6. Risk Guidance

LOW: No significant abnormality visible; continue routine monitoring.

MEDIUM: Possible degradation or minor abnormality; schedule inspection and monitor condition.

HIGH: Significant abnormality or combination of warning signs; prompt qualified inspection is required.

CRITICAL: Potential immediate safety or equipment hazard; follow emergency/shutdown procedures and escalate immediately.

7. AI-Assisted Inspection Guidance

Computer vision may identify visual indicators such as possible leakage, corrosion, damaged insulation, residue, or unusual surface conditions. These observations are decision-support evidence, not a definitive mechanical diagnosis. Confidence should be considered separately from risk severity.

8. Documentation

Every inspection should record the machine ID, date/time, observed condition, evidence source, recommended action, priority, and responsible maintenance team. Where applicable, reference the relevant SOP page or section.